Bilkent University

Department of Computer Engineering

Senior Design Project

Project Specification Report

Bil-Events : A mobile app to organize events in Bilkent University

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Specification Report
Feb 24, 2020

This report is submitted to the Department of Computer Engineering of Bilkent University in partial fulfillment of the requirements of the Senior Design Project course CS491/492.

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1.0 Introduction

According to YÖK's website, there are currently over 13.000 students studying in Bilkent [1]. Most of these students are members of some student clubs. These clubs host many different events for Bilkenters. However, there are several problems with the event notification system. First of all, since there is no unified notification system that people can access the information about events from various places. There are three main ways to access event information. One of them is from webmail notifications (BAIS), another one is from students clubs' posters and the third one is from SRS' GE 250/251 notifications. However, all of these three have their own flaws. To clarify what these flaws are actually, we should examine the event notification systems. The first one, the BAIS mail system, which sends over ten mail per day about not only events but also all notifications about Bilkent University and the seminars. It is hard to keep track of events and remember their dates. In addition to this, the BAIS mail system sends the emails unaesthetically, it causes resentment even for the students who want to invest their valuable time by attending the events. The second one, using paper posters is outdated, also a huge waste of paper and they are not well-distributed. They are so randomly distributed that it causes distraction among the students who want to find the right event for themselves. For instance, you can't find specific events posters in every building as they are not hung everywhere or they are overlapped when they are hung. For the students in the East Campus, this issue cannot even be mentioned because they cannot reach information about the events, because the students don't have any idea how to find the locations of the events and the events are generally held in the Main Campus, the distance between East Campus and the location of the events and lack of knowledge on how to find the location give resentment to the students who are in East Campus. BAIS e-mails and posters are not descriptive enough to handle that problem we mentioned. The last point we want to mention is that the update on SRS' event notification system does not even show the events that don't give GE 250/251 points. As a group of students who are in this system, we observed and suffered from these problems and as a solution we offer an app which will be accessed by everyone.

1.1 Description

Students in the Bilkent University are growth minded and they persistently attend the lectures, study and complete the projects in order to improve their skills and get themselves better. Outside of their academic responsibilities, they attend the events, learn new skills, engage in networking to find people with similar passions and goals. Events play a very important role in the students' vision. The strategy the Bilkent students prefer is to attend the events more, to network more and to get in touch with the people they have met in the events. That is why, instead of searching webmails, GE 250/251 page with their non-portable PCs which fails at fulfilling the tasks mentioned above or running around in order to find a poster about any event, accessing events and their details should be easy and fast. Bil-Events is an idea of a mobile application that offers an alternative solution to event notifications. It is an application that shows the currently existing clubs and their events with details. Using this app, users can easily access the events they desire with minimizing loss of their resources while searching the appropriate event. Therefore, we decided that it should be a mobile application to serve maximum amounts of people and increase the usability and accessibility. Not only events, but also seminars will be notified for students. Seminars are generally announced from Bilkent Webmail service but the same problem occurs which Bilkent Webmail is used for every notification and announcements. What we want is even to make the Bilkent Webmail service focus on one functionality which is to maintain the communication between Bilkent members without a waste of time. Bil-Events will also have a recommendation system which recommends students to several events based on their previous attended events. For example, if a student attended the "Sci-Fi" tagged event, in the future there is a high possibility that another event that holds "Sci-fi" tag will show up in the recommendation segment. In order to do that, recommendation for every user will have their own individual accounts that they will login with their Bilkent IDs, because we are aware that students in the Bilkent are unique in their own way.

1.2 Constraints

1.2.1 Implementation Constraints

- Bil-Event is an Android based application, so Java programming language will be used in the implementation step with Android studio. Github will be used to access and share the project work among the group members of the project.
- To embed a Recommendation Engine on our Android based application, we will use Apache PredictionIO, which is an open source Machine Learning Server built on top of a state-of-the-art open source stack for developers which is developed by Apache Mahout open-source project. PredictionIO has Java/ Android SDK and it has a well-written API Documentation and extensive tutorial on how to use the API [2].
- To make the Recommendation Engine work, we need to collect lots of data and store it in the database. PredictionIO has the dependencies on Java SDK version and database. We will use MongoDB to store the event data which is coming from students [3]. MongoDB is classified as a NoSQL database program and it is flexible, it stores in flexible, JSON-like documents.[4] After collecting lots of data and storing on MongoDB or having lots of data and storing the data using Big Data Technologies such as Spark and Hadoop, we can use PredictionIO's own Event Server which collects data via a RESTful API. Event Server uses Apache HBase as a data store [5].

1.2.2 Sustainability Constraints

- For sustainability constraints, if there will be no change in the application distribution environment (Google Play Store) our application will continue to use these environments. However Bil-Events may not be working in lower Android versions.
- Our recommender engine PredictionIO has various dependencies such as Elasticsearch and HBase, they could stop running or they could not run properly or we could encounter the problem of increasing JVM heap size of

the Event Server or increasing heap space memory for deploying the recommendation engine [6].

- Bil-Events application will be updated regularly in order to fix the bugs if any happens. In addition to this, the recommendation engine will be checked to ensure it is up and running properly and backup key will be used to prevent losing data caused by database failure [7].
- Improvements and new features might be added in the application based on users feedback.

1.2.3 Privacy & Security Constraints

- Every user will have a private account. Each account will have a unique username which is the student ID and a password.
- Our application will not store personal data such as social security numbers, TC identification numbers, credit card numbers and so on.
- The students' personal data cannot be accessed by anyone. It will be encrypted and MongoDB supports the encryption via Encrypted Storage Engine [8]

1.2.4 Time Constraints

- Our application will be finished on the 24th of December 2020.
- We plan to finish the implementation stage as soon as possible to test our application. Since our application will have multiple parts in the system, we need to make sure that they will be running collaboratively and problem-freely.

1.2.5 Social Constraints

- Bil-Events will only serve Bilkent students. Therefore it will be only available for Bilkent Students. If a person is not a part of the university, that person will not be able to use the application.
- In the future, if Bilkent allows people that have no connection with Bilkent, enter the club events, foreign users can benefit from Bil-Event.

1.2.6 Language Constraints

- Our application's language will be English. Since, English is considered as one of the most dominant global languages.
- Depending on demand, new language may be added to the application.

1.2.7 Technological Constraints

- Our application Bil-Events will be mobile only application.
- It will be accessible in only Google Play Store. After the initial application starts it may be launched in the Apple Store as well.
- Based on availability of the users and the size of data we will obtain, we could consider using Cloud Storage such as AWS or Google Cloud and Big Data Technologies such as Spark or Hadoop.

1.3 Professional and Ethical Issues

1.3.1 Ethical Issues

- Bil-Events application will not collect personal information such as contact number, credit card information, TC identification number. In the case of storing other personal data, we will encrypt the data and make sure that the data is secure.

1.3.2 Professional Issues

- Source code of the application will be stored as a private GitHub code.
- Project process will be distributed equally for each member.
- Each member is responsible for helping other members in a place they lack knowledge. In this way both sides will learn how to work collaboratively.
- Each member is responsible for making extensive research about which technology they will use and whether the technology that the member will use is compatible with the technologies which the remaining members will use.
- Each member is responsible for keeping in touch with the supervisor, for making a respectable amount of improvements on each week and for sharing the improvements with the supervisor.

- Each member is responsible for getting advice from the supervisor if the other members have the lack of knowledge about the problem that a specific member is having.
- We will meet up with each other at least once a week. This meeting may be face-to-face meetings or on a digital platform such as “Skype” or “Discord”. We will meet up with our supervisor regularly, if he is available.

2.0 Requirements

2.1 Functional Requirements

- Users will create accounts with their student ID’s. The users which are not in the Student Database Management System of the Bilkent University cannot create an account.
- Users will be asked to specify their areas of interests (e.g. art, music, etc) that will be used for suggesting events in the recommendation process.
- The users can search for the events and seminars in Bilkent.
- The user can select an event to see details such as description, location, starting time etc.
- Bil-Event will list events in a way that specific events which are recommended for the user will be on top.
- The user can choose to attend an event.
- The user can choose to create an event but to create an event, the user must be the official club leader.
- After the event is posted on the app, everyone can access it immediately.
- After attending an event, the user can rate the event and write reviews about his/her experiences.
- The user can rate attended events on multiple categories.

2.2 Non-Functional Requirements

2.2.1 Usability

The application will be easy to use for both organizers and participants. Therefore, the user interface must be simple enough for both users. UI/UX Design are taken into account when we develop our application.

2.2.2 Reliability

As events in Bilkent University will be added by club leaders, the events are expected to happen. As the seminar is managed by the collaboration between student club leaders and Bilkent University , events will be reliable. The rating of events and the reviews about the events are expected to increase the reliability of the events.

2.2.3 Extensibility

The application will be easy to add features (e.g. idea of chat, friend requests etc). Hence, it should be easy to update with respect to both its background implementation and user interface design. The requirements of the users are going to be taken into account when adding new features.

2.2.4 Privacy

User information will be private. Unless taking permission of the user, user information should not be accessed by anyone. Also in the application, if participants want to hide their information (event history, location, age, interests, and etc.), they should be given this option. The collected data from users are encrypted and protected with the backup and encryption keys. Selling the user data to the third parties are prohibited regardless of the consent of users.

2.2.5 Flexibility

The application will be suitable for all Bilkent students, even preparatory students. It should also provide online features regarding the different user expectations.

2.2.6 Portability and Accessibility

As the product is a mobile application, it will be portable and accessible for all Bilkent students from everywhere. Internet connection is needed. The app won't require high quality hardware. We could consider adding some accessibility features to our disabled users.

2.2.7 Sustainability

Since the highschool students are applying to Bilkent University every year and students are actively participating in the events of Student Clubs because of their interest in the club or GE 250/251, the application preserves its sustainability.

3.0 References

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